
VideoGenDoctor: Structured Failure Diagnosis, Evidence Localization, and Repair Planning for Controllable Video Generation

Anonymous Author(s)

Affiliation

Address

email

Abstract

Controllable video generation increasingly relies on structured conditions such as character identity, camera trajectory, required props, temporal actions, and multi-shot specifications. Existing evaluation pipelines mostly return scalar quality or alignment scores, which are insufficient for debugging because they rarely identify what failed, where it failed, which evidence supports the diagnosis, or how the next generation attempt should be repaired.

We present VideoGenDoctor, a structured framework for failure diagnosis, evidence localization, and repair planning. Given a generated video and its prompt or structured specification, VideoGenDoctor converts failures into code-span-target-plan records grounded in temporal evidence and representative keyframes. The default implementation combines lightweight feature-based screening, optional VLM verification, and a patch compiler that maps localized failures to repair plans.

We position VideoGenDoctor-Bench-v0 as a controlled diagnostic fixture rather than a broad open-world benchmark. It contains 324 videos, 9 deterministic perturbation types, and 2,016 human-verified evidence spans covering 12 observed failure codes from a 38-code taxonomy. On this fixture, VideoGenDoctor-diagnosis reaches macro-F1 0.9110 and tIoU@0.5 0.7316; Rule+GPT-4V reaches 0.9276 and 0.7688. In automatic closed-loop evaluation, VideoGenDoctor-full improves Pass@1/2 from 24.07%/35.80% for Score-only feedback to 64.81%/83.64%. Rule+GPT-4V + repair loop remains the strongest absolute scorer on the controlled fixture, but at $2.75\times$ relative cost; we therefore treat VideoGenDoctor-full as the default operating point and use the GPT-4V front end as a high-cost reference. On a 240-video failure-enriched subset from four controllable video generators, the default VideoGenDoctor pipeline improves front-end diagnosis macro-F1 from 0.7992 to 0.8264 and tIoU@0.5 from 0.5431 to 0.5982 over direct structured VLM reporting, while also improving downstream repair utility under the same repair protocol. These results support structured, auditable repair records as a practical interface for controlled diagnosis and repair planning, while leaving broad open-world generality as future validation.

1 Introduction

Recent diffusion-based models can synthesize multi-second videos conditioned on text, reference images, camera trajectories, character identities, and structured shot scripts Yang et al. [2024], Blattmann et al. [2023], Wang et al. [2023], Yuan et al. [2024]. In production-style workflows,

ShotIR-like specifications define per-shot props, characters, camera movement, shot type, and action order. When a generated video violates such specifications, however, users still lack a systematic interface for locating the failure and deciding the next intervention.

Current video-generation evaluators such as VBench Huang et al. [2024], EvalCrafter Liu et al. [2024], VideoScore He et al. [2024], and T2V-CompBench Sun et al. [2025] report aggregate quality, motion, or alignment scores. These scores are useful for model comparison but weak as debugging signals: a low score does not indicate whether the character identity drifted, a required prop disappeared, camera motion was violated, or a specific temporal segment should be regenerated.

VideoGenDoctor reframes controllable-video evaluation as structured diagnosis and repair planning. Given a video V and specification S , it outputs records

$$R = \{(c_i, t_0^i, t_1^i, y_i, \sigma_i, K_i, T_i, a_i)\},$$

where c_i is a failure code, (t_0^i, t_1^i) is the evidence span, y_i is the affected target, σ_i is confidence, K_i contains representative keyframes, T_i stores optional track-level evidence, and a_i is a compiled repair action. The interface is designed to be machine-readable and human-auditable; executable repair depends on the capabilities exposed by the downstream generator or editor.

Scope. We separate the interface claim from the open-world generality claim. The expanded controlled fixture supports the claim that code–span–target–plan records provide auditable diagnosis and useful repair plans under known perturbations. A failure-enriched real-generator subset suggests transfer beyond deterministic perturbations, but broader coverage across generators, domains, styles, and long-form videos remains future work. We therefore call VideoGenDoctor-Bench-v0 a controlled diagnostic fixture rather than a general open-world benchmark.

Contributions. This paper contributes: (1) a code–span–target–plan representation for controllable video debugging; (2) a modular framework combining feature-based candidate generation, optional VLM verification, evidence aggregation, and patch compilation; (3) VideoGenDoctor-Bench-v0, with 324 controlled videos, 2,016 verified evidence spans, 9 perturbation types, and 12 observed codes, plus a 240-video failure-enriched real-generator subset covering 18 observed codes; and (4) an evaluation against direct VLM diagnosis, VLM-to-patch repair, blind human repair validation, per-code reliability analysis, paired bootstrap comparisons, and transfer checks on naturally occurring failures.

2 Related Work

Video generation evaluation. Existing benchmarks and evaluators measure aggregate video quality, motion smoothness, subject consistency, and text-video alignment Huang et al. [2024], Liu et al. [2024], He et al. [2024], Sun et al. [2025]. They are valuable for ranking models but do not usually produce localized evidence or executable repair plans. VideoGenDoctor instead returns structured failure records tied to temporal spans and repair templates. A compact capability comparison is provided in Appendix A, Table 9.

VLM critics and temporal grounding. Large vision-language models can inspect sampled frames and produce critiques Liu et al. [2023], Chen et al. [2024], OpenAI [2023]. We therefore treat direct VLM reporting and VLM-to-patch generation as strong baselines rather than strawmen. The central question is whether constraining critique through code–span–target records improves temporal grounding, schema validity, auditability, repair executability, and cost. For temporal localization, we adapt temporal intersection-over-union:

$$\text{tIoU} = \frac{\max(0, \min(\hat{t}_1, t_1^*) - \max(\hat{t}_0, t_0^*))}{\max(\hat{t}_1, t_1^*) - \min(\hat{t}_0, t_0^*)}.$$

Automated repair. Program repair systems typically localize a defect, classify it, and generate a patch. VideoGenDoctor adapts this pattern to controllable video generation: each diagnosis must expose evidence and map to a supported repair family, connecting evaluation directly to iterative generation.

79 3 VideoGenDoctor

80 3.1 Overview

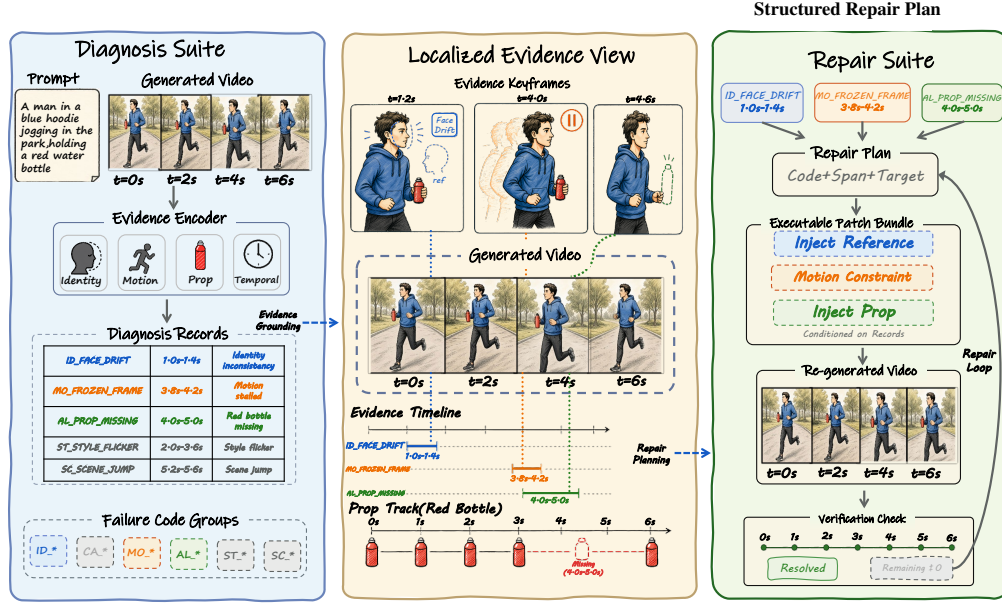


Figure 1: VideoGenDoctor framework overview. The system converts a generated video and its prompt or structured specification into failure-as-code diagnosis records, grounds each record in localized temporal evidence, maps records to structured repair plans, and verifies regenerated videos in a closed loop.

81 VideoGenDoctor contains four modules: segmentation and feature extraction, candidate diagnosis,
 82 optional VLM verification, and patch compilation. The first module samples keyframes and computes
 83 lightweight signals; the second applies thresholded rules to produce failure candidates; the third
 84 calibrates top-ranked candidates using structured VLM questions; and the fourth maps retained
 85 diagnosis records to structured repair actions such as reference injection, prop injection, camera
 86 override, segment regeneration, or style correction.

87 3.2 Failure Records and Patch Templates

88 VideoGenDoctor uses a machine-readable taxonomy with 38 codes grouped into Identity, Camera,
 89 Motion, Alignment, Style, and Scene. The current controlled fixture evaluates 12 observed codes; the
 90 real-generator subset covers additional naturally occurring codes. Codes outside these observed sets
 91 define an extensible namespace and are not treated as empirically validated. The full taxonomy and
 92 group-level description are provided in Appendix A and Appendix C.

93 Each predicted failure is represented as

$$r = (c, t_0, t_1, y, \tilde{\sigma}, K, T, a),$$

94 where c is the taxonomy code, (t_0, t_1) is the evidence span, y is the affected target, $\tilde{\sigma}$ is confidence,
 95 K stores representative keyframes, T stores optional tracks, and a is the compiled repair action.
 96 Detectors and judges can be replaced without changing this schema.

Table 1: Representative failure types and repair-plan template mappings. A diagnosis record combines a failure type with a localized span, target, confidence, evidence, and repair-plan action.

Failure type	Evidence signal	Repair action
Face identity drift	Face embedding drift across keyframes	Reference-frame injection
Body / clothing drift	Character-region appearance drift	Reference-frame or character-anchor injection
Camera movement deviation	Camera flow trace or trajectory mismatch	Camera movement override
Frozen frames or action stalls	Near-zero flow or stalled pose track	Regenerate affected segment
Missing required prop	Object presence check or VLM prop-absence judgment	Prop injection
Style inconsistency	Style or color embedding drift across segments	Style correction
Background inconsistency	Scene/background embedding drift	Background anchoring

3.3 Candidate Diagnosis and Verification

Given video V , we partition it into 2-second fixed-stride segments and sample six keyframes per segment. The default implementation computes semantic embedding drift with OpenCLIP Cherti et al. [2023], optical flow with RAFT Teed and Deng [2020] or OpenCV Farneback Bradski [2000], Farneback [2003], face drift with InsightFace Deng et al. [2019], and object or prop cues with YOLOv8 Ultralytics [2023] when enabled. For each segment s and failure code c , rule ρ_c produces a score $\sigma_{s,c} \in [0, 1]$; top-ranked candidates inherit segment boundaries and keyframes ranked by frame-level anomaly scores.

The optional Stage-2 judge receives the top- K candidate segments ($K = 3$ by default), sampled keyframes, temporal bounds, candidate codes, and relevant prompt or ShotIR fields. It answers structured questions rather than writing an unconstrained report. Final confidence is

$$\tilde{\sigma}_{s,c} = (1 - \alpha)\sigma_{s,c}^{(1)} + \alpha\sigma_{s,c}^{(2)},$$

with $\alpha = 0.6$ by default. Hosted endpoints, prompts, sampled frames, raw responses, parsed JSON, and latency metadata are logged for reproducibility.

3.4 Evidence Localization and Repair Planning

Each retained record stores an evidence object $E = (t_0, t_1, K, T)$ so that the diagnosis can be audited. Fixed segments are sufficient for the current fixture; future versions can refine spans using frame-level peaks and track-based merging. The patch compiler then emits either ShotIR field-level diffs or generator-agnostic repair plans. Each plan preserves the triggering code, evidence span, affected target, and rationale. In closed-loop evaluation, a video passes when all failure confidences fall below $\tau = 0.5$ or the loop reaches $K_{\max} = 3$; because this automatic criterion is verifier-dependent, Section 4.5 also reports blind human repair validation.

4 Experiments

4.1 Benchmark and Evaluation Protocol

We evaluate VideoGenDoctor as a diagnosis-and-repair system. The main questions are whether predicted failure codes are correct, evidence spans are temporally localized, repair plans are valid and useful, and repaired videos satisfy the specification under independent human judgment.

Controlled fixture. VideoGenDoctor-Bench-v0 contains 324 perturbed videos, 2,016 verified evidence spans, 9 deterministic perturbation types, and 12 observed failure codes. It is built from 18 clean source clips across six domain groups; each clip is paired with a ShotIR-style specification and perturbed by nine deterministic operators under two seeds. The fixture validates interface consistency, evidence auditability, and repair-planning usefulness under controlled failures; it is not intended as a comprehensive open-world benchmark. Construction details and operator-to-template alignment are moved to Appendix A, Table 11, and Appendix B, Table 14.

Real-failure and real-normal subsets. We collect 240 naturally occurring failure videos from CogVideoX, Wan, Stable Video Diffusion, and HunyuanVideo Yang et al. [2024], Wan-AI [2025], Blattmann et al. [2023], Tencent Hunyuan Foundation Model Team [2025]. The subset is failure-enriched: from 480 raw generations, we retain videos only when annotators verify at least one visible specification violation. Sampling is balanced across the four generators at 60 retained failure videos each, with generator-specific input adapters logged in a per-video JSON manifest: CogVideoX and Wan use ShotIR-to-prompt text templates, Stable Video Diffusion uses reference-frame conditioning, and HunyuanVideo uses a multi-shot prompt template. The manifest records prompt or spec identifiers, converted inputs, generation settings, screening status, verified codes, and evidence spans so that the transfer check remains auditable even though it is not a prevalence estimate. We also retain 120 real-normal videos from the same raw generations to estimate false positives and unnecessary repair behavior. Ambiguous and low-quality exclusions are treated as stress analyses rather than standard accuracy benchmarks. Detailed source composition, normal-set behavior, temporal evidence profiles, and stress tables are reported in Appendix B.

Annotation. A reliability subset of 120 videos and 768 candidate evidence spans is dual-annotated before adjudication. We report code-level Cohen’s κ and span-level annotator-to-annotator tIoU in Appendix B, Table 20. Agreement is substantial for failure codes and lower for exact temporal boundaries, reflecting the ambiguity of fine-grained evidence spans.

Table 2: Dataset statistics. The first row is the expanded controlled perturbation fixture. The second row is the expanded real-failure subset collected from naturally occurring failures across multiple generators.

Split	#Vid	#Evidence	AvgDur	PerturbTypes	#Codes
Controlled fixture	324	2016	10.46s	9	12
Real-failure subset	240	1536	8.93s	–	18

4.2 Compared Methods and Metrics

We compare internal diagnosis variants (Prior-only, Rule-only, Rule+DummyJudge, Rule+Open-VLM, Rule+GPT-4V, and VideoGenDoctor-diagnosis), repair variants (VideoGenDoctor-patch, Patch+Judge, and VideoGenDoctor-full), and external VLM baselines including free-form VLM reporting, structured VLM reporting, self-consistency, VLM temporal proposal + patch, VLM-to-patch, and Score+LLM-to-patch. Most direct VLM baselines share the same hosted GPT-4V endpoint as Rule+GPT-4V to isolate prompt structure, output schema, and repair-planning protocol rather than model backbone; Rule+Open-VLM is included as an open-model reference. Whenever a table reports human repair success for a diagnosis-oriented front end, that front end is paired with the same downstream repair loop so that only the diagnostic interface changes. Implementation details are reported in Appendix B, Table 21; strengthened VLM-agent results are summarized in Appendix A, Table 12.

We report macro-F1 and micro-F1 for failure-code detection, tIoU@0.3/0.5 and Top- K keyframe hit rate for localization, automatic and human Pass@1/2 for repair, patch usefulness, new-artifact rate, and video-level bootstrap 95% confidence intervals. All span-aware metrics use the same matching protocol: records are grouped by failure code, target-incompatible predictions are filtered, and remaining pairs are matched by Hungarian assignment Kuhn [1955] using tIoU as the primary score and confidence only as a tie-breaker. Invalid or missing spans receive no localization credit.

4.3 Failure-Code Detection

Table 3: Failure-code detection on the expanded controlled fixture. Direct VLM baselines are included to compare VideoGenDoctor with strong agent-style alternatives. Confidence intervals are estimated by video-level bootstrap resampling. Best values are boldfaced.

Method	Macro-F1	Micro-F1	95% CI
Prior-only	0.4479	0.4635	[0.419, 0.477]
Rule+DummyJudge	0.8584	0.8716	[0.836, 0.879]
Rule-only	0.8926	0.9038	[0.872, 0.911]
Rule+Open-VLM	0.9047	0.9175	[0.885, 0.922]
VideoGenDoctor-diagnosis	0.9110	0.9243	[0.893, 0.928]
Rule+GPT-4V	0.9276	0.9386	[0.912, 0.941]
VLM free-form report	0.8402	0.8627	[0.814, 0.866]
VLM structured report	0.8869	0.9018	[0.864, 0.908]

VideoGenDoctor-diagnosis outperforms free-form and structured direct VLM reporting on macro-F1, while Rule+GPT-4V remains the strongest configuration. The larger gains in the localization results below indicate that the main advantage of structured diagnosis is not only detecting that a failure exists, but grounding it stably enough to support repair.

4.4 Per-Code Reliability and Evidence Localization

The controlled fixture covers twelve observed codes. Detailed per-code precision, recall, F1, and tIoU are reported in Appendix B, Table 22; the confusion matrix below summarizes dominant error modes.

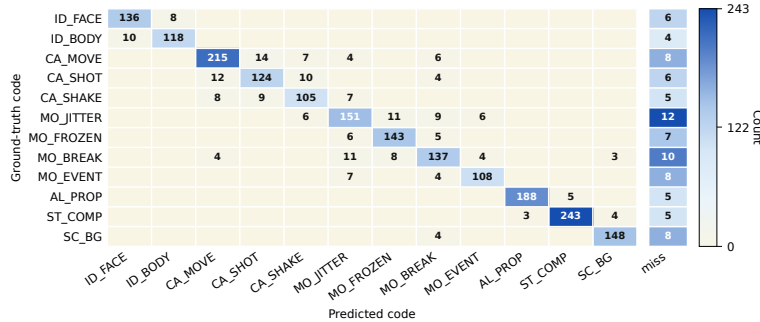


Figure 2: Per-code confusion matrix for the 12 observed taxonomy codes on the expanded controlled fixture. The strongest off-diagonal confusions occur among camera and motion failures, while identity, prop-missing, compression-artifact, and scene-background failures remain comparatively concentrated on the diagonal.

Table 4: Evidence localization against human evidence spans on the expanded controlled fixture. Confidence intervals are estimated by video-level bootstrap resampling. Best values are boldfaced.

Method	tIoU@0.3	tIoU@0.5	Top-1	Top-3	95% CI
Prior-only	0.5817	0.4295	0.1188	0.3315	[0.403, 0.457]
Rule+DummyJudge	0.7282	0.6128	0.2241	0.4259	[0.586, 0.639]
Rule-only	0.7836	0.6714	0.2670	0.5123	[0.645, 0.698]
Rule+Open-VLM	0.8089	0.7062	0.3565	0.5941	[0.681, 0.731]
VideoGenDoctor-diagnosis	0.8261	0.7316	0.4336	0.6815	[0.707, 0.754]
Rule+GPT-4V	0.8554	0.7688	0.4923	0.7747	[0.746, 0.790]
VLM free-form report	0.7204	0.5986	0.2994	0.5117	[0.570, 0.625]
VLM structured report	0.7812	0.6639	0.3651	0.5858	[0.637, 0.690]

Table 5: Closed-loop repair evaluation on the expanded controlled fixture. Automatic metrics are computed on all 324 controlled videos. Human metrics are computed on a stratified blind-evaluation subset of 144 videos. Best pass rates and patch-usefulness values are boldfaced; the new-artifact rate is reported for context rather than bold-ranked because abstention-heavy baselines are not directly comparable.

Method	Auto P@1	95% CI	Auto P@2	95% CI	Human Pass@1	95% CI	Human Pass@2	95% CI	Patch useful	New artifacts
Score-only	0.2407	[0.196, 0.291]	0.3580	[0.307, 0.411]	0.1875	[0.130, 0.257]	0.2778	[0.207, 0.355]	2.28	0.0833
Patch-only	0.5247	[0.471, 0.579]	0.6975	[0.646, 0.744]	0.4792	[0.399, 0.559]	0.6319	[0.551, 0.706]	3.50	0.1736
VLM-to-patch	0.5617	[0.507, 0.615]	0.7438	[0.694, 0.788]	0.5069	[0.427, 0.586]	0.6736	[0.594, 0.746]	3.64	0.2153
Patch+Judge	0.6296	[0.575, 0.681]	0.8179	[0.773, 0.857]	0.5625	[0.482, 0.640]	0.7431	[0.667, 0.807]	3.93	0.1458
VideoGenDoctor-full	0.6481	[0.594, 0.699]	0.8364	[0.793, 0.873]	0.5833	[0.503, 0.660]	0.7639	[0.689, 0.826]	4.11	0.1319

4.5 Closed-Loop Repair and Human Validation

Closed-loop repair is evaluated on the controlled fixture. Automatic Pass@1/2 is computed on all 324 videos using the system verifier, while blind human Pass@1/2 is computed on a stratified 144-video subset. Human raters see the specification and output video but not method names or diagnosis records, so the human metric is the primary check against verifier-coupled gains.

The main repair gain is most consistent with structured repair planning and verification rather than with the final loop wrapper alone. The paired comparison in Section 4.7 shows that the difference between Patch+Judge and VideoGenDoctor-full is small and not statistically decisive, so the full loop is best interpreted as traceable workflow integration.

4.6 Controlled Fixture versus Real Generator Failures

Table 6: Generalization check on expanded controlled perturbations and naturally occurring generator failures. Macro-F1 and tIoU@0.5 are computed from each method’s diagnosis front end; Patch useful and Human Pass@2 evaluate the corresponding downstream repair protocol. Best values within each subset are boldfaced.

Subset	Method	Macro-F1	tIoU@0.5	Patch useful	Human Pass@2
Controlled fixture	VLM structured report + repair loop	0.8869	0.6639	3.62	0.6810
Controlled fixture	VideoGenDoctor default pipeline	0.9110	0.7316	4.11	0.7639
Real-failure subset	VLM structured report + repair loop	0.7992	0.5431	3.36	0.6075
Real-failure subset	VideoGenDoctor default pipeline	0.8264	0.5982	3.82	0.6908

All methods degrade on naturally occurring failures, confirming that the controlled fixture is not a substitute for open-world evaluation. The real-failure subset contains more overlapping artifacts, less isolated evidence, and weaker alignment between specification fields and visible deviations. Because this subset is failure-enriched and balanced by construction, these numbers measure transfer under verified failures rather than real-world prevalence. Per-code and per-generator results are reported in Appendix B, Tables 23 and 24.

4.7 Ablation and Generalization Checks

Table 7: Repair ablation on the expanded controlled fixture. We isolate the contribution of failure codes, temporal spans, evidence keyframes, patch templates, and verification. Best utility values are boldfaced; the new-artifact rate is reported for context rather than bold-ranked.

Repair setting	Code	Target	Span	Evidence	Template	Human Pass@2	Patch useful	New artifacts
Score-only	✗	✗	✗	✗	✗	0.2778	2.28	0.0833
Code-only patch	✓	✗	✗	✗	✓	0.4583	2.91	0.1597
Code+target patch	✓	✓	✗	✗	✓	0.5417	3.24	0.1736
Code+span patch	✓	✗	✓	✗	✓	0.6250	3.57	0.1667
Code+span+evidence patch	✓	✗	✓	✓	✓	0.6875	3.80	0.1528
VLM-to-patch	optional	optional	optional	optional	✗	0.6736	3.64	0.2153
Patch+Judge	✓	✓	✓	✓	✓	0.7431	3.93	0.1458
VideoGenDoctor-full	✓	✓	✓	✓	✓	0.7639	4.11	0.1319

The ablation tests whether localized evidence and template-constrained repair contribute beyond code prediction alone under a shared backend. The improvement from code-only to code+span and

code+span+evidence settings supports the claim that evidence localization is operationally useful for repair planning, but it does not by itself identify a single causal component because target fields, span quality, and backend executability interact in the repair loop.

Table 8: Paired comparison of repair variants. We report paired bootstrap differences in blind human Pass@2. Small differences between Patch+Judge and Full loop should be interpreted cautiously.

Comparison	Δ Human Pass@2	95% CI	Interpretation
Patch-only vs Score-only	0.3541	[0.250, 0.458]	substantial gain
VLM-to-patch vs Patch-only	0.0417	[-0.054, 0.139]	small / not decisive
Patch+Judge vs Patch-only	0.1112	[0.021, 0.215]	verification helps
VideoGenDoctor-full vs Patch+Judge	0.0208	[-0.063, 0.104]	small / not decisive

We also perform leave-one-perturbation-out and leave-one-source-group-out evaluations. Performance drops under held-out perturbations, especially for missing-event, camera-shake, and temporal-discontinuity operators, but remains above prior-only and score-only baselines. These results suggest that the system is not merely memorizing perturbation templates, while also confirming that unseen-operator generalization is weaker than in-distribution controlled evaluation. Detailed tables and the cost-performance trade-off are reported in Appendix B, Tables 25, 26, and Appendix A, Table 13.

5 Discussion

The current evidence supports a scoped claim: structured code–span–target–plan records improve diagnosis and repair planning over score-only feedback and direct VLM reporting on the controlled fixture and the current failure-enriched real-generator subset. It does not show that VideoGenDoctor solves open-world video repair. The controlled fixture demonstrates auditable localization under known perturbations, while the real-failure subset provides an initial but still limited transfer check.

The value of VideoGenDoctor should be assessed as an interface contribution rather than as a new video generation model or a learned repair algorithm. Its advantages lie in schema validity, temporal grounding, repair-plan usefulness, human auditability, and cost-performance. Rule+GPT-4V + repair loop is the strongest absolute scorer in the current controlled evaluation, but it relies on a more expensive hosted front end and is best interpreted as an upper-reference configuration rather than the default operating point. We therefore headline VideoGenDoctor-full because it preserves the structured interface and audit trail while delivering the main closed-loop gain at the paper’s default cost point. The modular design also allows feature extractors, rules, VLM judges, and patch compilers to be replaced independently; Stage-2 judging is restricted to top-ranked candidate segments, and the closed loop bounds repair by $K_{\max}$.

Limitations. The expanded fixture remains controlled; the real-failure subset is failure-enriched and cannot estimate natural failure prevalence; automatic pass rates are partly verifier-dependent; only observed taxonomy codes support empirical claims; and repair assumes access to a generator or editor that can consume at least part of the patch vocabulary. Broader validation requires larger generator coverage, longer videos, more domains and styles, stronger adapter executability tests, and larger multi-annotator studies.

6 Conclusion

We presented VideoGenDoctor, a structured framework for failure diagnosis, evidence localization, and repair planning in controllable video generation. The core contribution is an auditable code–span–target–plan interface that connects evaluation to repair planning. The expanded evaluation consists of a 324-video controlled diagnostic fixture with 2,016 verified evidence spans and a 240-video failure-enriched real-generator subset with 1,536 verified evidence spans. The controlled fixture covers 12 observed taxonomy codes under 9 deterministic perturbation operators, while the real-generator subset covers 18 observed taxonomy codes across four publicly identifiable video generation systems. The primary empirical gain comes from structured repair planning and independent verification, while the full closed loop mainly provides traceable integration from diagnosis to evidence, patch planning,

234 re-rendering, and verification. The broader generality claim remains scoped by the controlled and
235 failure-enriched nature of the current evaluation sets.

236 7 Ethics Statement

237 VideoGenDoctor is a diagnostic tool intended to improve the quality and controllability of generated
238 videos. Any tool that improves generation reliability could also be misused to strengthen misleading
239 media. In sensitive deployments, automated repair should be paired with provenance tracking,
240 watermarking, human review, and access control. Evidence localization may surface keyframes
241 containing sensitive information, so downstream systems should support local processing, keyframe
242 redaction, coarse time-binning, and privacy-preserving aggregation where appropriate. The taxonomy
243 also encodes normative judgments about what counts as a failure; creative workflows should expose
244 thresholds and permit human override.

245 8 Reproducibility and Release

246 For artifact review, the planned release package includes the reference implementation, report
247 schema, taxonomy, patch compiler, controlled fixture generation scripts, annotations, prediction logs,
248 evaluation scripts, prompt templates for VLM baselines, and scripts for bootstrap confidence intervals.
249 Each run is expected to record configuration files, model versions, package versions, random seeds,
250 prompts, raw VLM responses, and output paths. The intended artifact should permit CPU-only
251 recomputation of tables from serialized predictions and annotations in seconds; full feature extraction
252 and VLM judging remain configuration-dependent.

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293 A Supplementary Main-Text Results

294 This appendix stores tables and figures moved out of the main text to keep the core narrative compact
295 while preserving the audit trail for comparisons, taxonomy scope, fixture construction, temporal
296 evidence profiles, strengthened VLM baselines, repair visualization, and cost-performance trade-offs.

Table 9: Capability comparison with related evaluation and critique paradigms. Here “Global score” denotes scalar quality, alignment, or verifier-style pass outputs rather than localized evidence records. Direct VLM reporting is treated as a strong baseline rather than only comparing against score-only benchmarks.

System	Global score	Code	Evidence	Patch	Closed-Loop
VBench	✓	✗	✗	✗	✗
EvalCrafter	✓	✗	✗	✗	✗
VideoScore	✓	✗	✗	✗	✗
T2V-CompBench	✓	partial	✗	✗	✗
Direct VLM report	partial	partial	partial	partial	✗
VideoGenDoctor	✓	✓	✓	✓	✓

Table 10: Failure-as-code taxonomy groups. Each group defines a namespace for diagnosis records and links to evidence procedures and repair-plan templates.

Group	Operational scope
Identity	Identity drift, face/body consistency, wrong character, clothing consistency.
Camera	Shot type, camera movement, viewpoint angle, unintended zoom, camera shake.
Motion	Action timing, frame continuity, motion physics, frozen frames, segment breaks.
Alignment	Prompt adherence, required objects and props, character count, prop placement.
Style	Color consistency, texture flicker, compression artifacts, art-style consistency.
Scene	Background consistency, lighting, environment match, scene-object stability.

Table 11: Composition of the expanded controlled diagnostic fixture. Each domain group contains three clean source clips. Every source clip is paired with a ShotIR-style specification and perturbed by nine deterministic operators under two random seeds.

Domain group	#Source clips	Avg. duration	Resolution	#Perturb. ops	#Seeds	#Videos
Indoor / human-object	3	10.1s	768×432	9	2	54
Outdoor / motion	3	10.8s	768×432	9	2	54
Multi-object interaction	3	10.4s	768×432	9	2	54
Camera-control scene	3	11.0s	768×432	9	2	54
Style / lighting variation	3	10.2s	768×432	9	2	54
Scene-change / background	3	10.3s	768×432	9	2	54
Total	18	10.46s	—	9	2	324

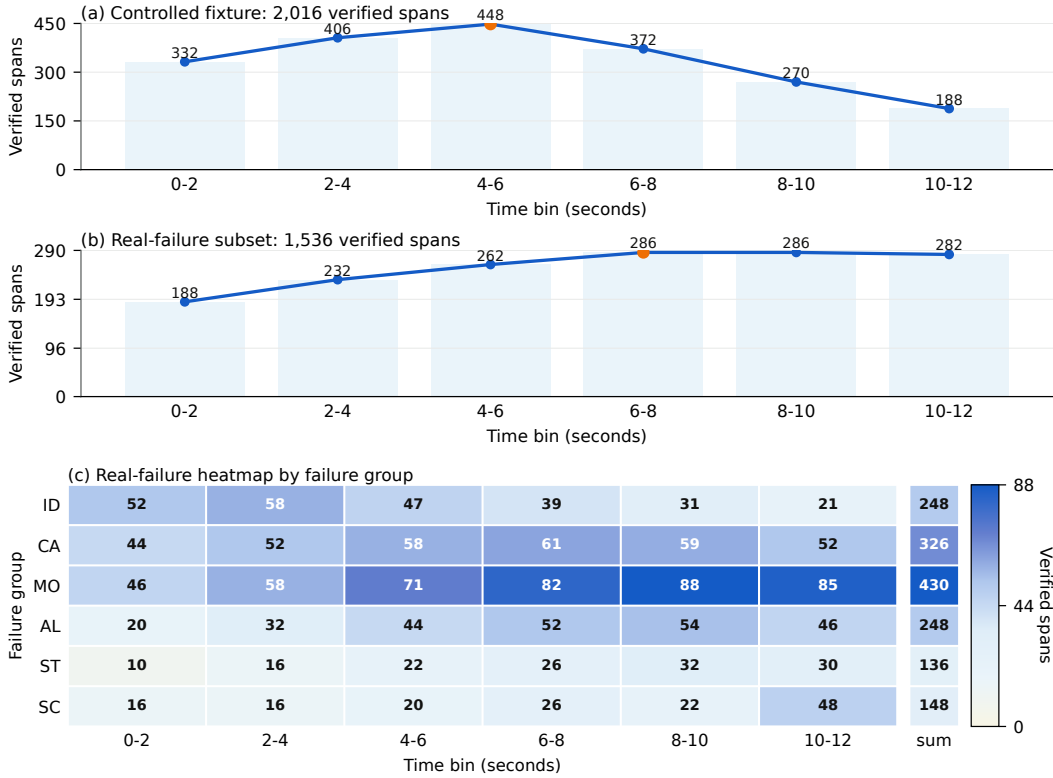


Figure 3: Temporal distribution of verified evidence spans in the expanded evaluation sets. Panel (a) aggregates the 2,016 controlled-fixture spans over 2-second bins and shows the front- and mid-loaded pattern expected from deterministic perturbation operators. Panel (b) summarizes the 1,536 real-failure spans over the same time bins, making the later and more diffuse temporal mass directly comparable to panel (a). Panel (c) further breaks the real-failure spans down by failure group and time bin, highlighting that motion and camera failures dominate the later intervals and overlap more strongly than in the controlled fixture.

Table 12: Strengthened VLM-agent baselines on the expanded controlled fixture. All rows are evaluated under the same downstream repair protocol; diagnosis-only front ends are explicitly paired with that shared repair loop when human repair success is reported. Self-consistency improves direct VLM reporting but increases cost. The validity column reports whether the emitted action stays inside the supported repair-plan vocabulary; it does not claim generator-adaptor executability. Best values are boldfaced; higher is better except for relative cost, where lower is better.

Method	Macro-F1	tIoU@0.5	Human Pass@2	Schema validity	Patch vocabulary validity	Relative cost
VLM free-form report	0.8402	0.5986	0.6125	0.742	0.604	1.12×
VLM structured report	0.8869	0.6639	0.6810	0.914	0.732	1.16×
VLM structured + self-consistency	0.8978	0.6824	0.7042	0.951	0.755	3.48×
VLM temporal proposal + patch	0.8915	0.7018	0.7076	0.936	0.802	2.05×
VideoGenDoctor-full	0.9110	0.7316	0.7639	0.991	0.953	1.00×
Rule+GPT-4V + repair loop	0.9276	0.7688	0.7917	0.993	0.957	2.75×

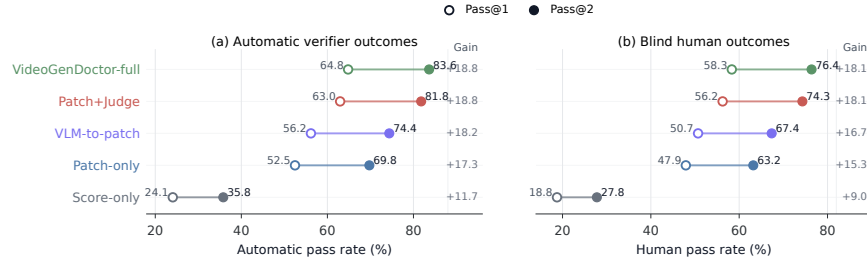


Figure 4: Closed-loop repair comparison on the expanded controlled fixture. Left: automatic verifier Pass@1/2 on all 324 controlled videos. Right: blind human Pass@1/2 on the stratified 144-video human-evaluation subset. Automatic pass rates may be coupled to the system verifier, while human pass rates provide an independent check of repair success.

Table 13: Cost-performance trade-off under a shared downstream repair protocol. Diagnosis-only front ends are paired with the same repair loop before reporting Human Pass@2. Cost is reported relative to the default VideoGenDoctor-full configuration. Values in the last two columns are multiplicative factors relative to VideoGenDoctor-full (1.00×). Best values are boldfaced; higher is better for task metrics, lower is better for cost and latency.

Method	Macro-F1	tIoU@0.5	Human Pass@2	Relative cost	Relative latency
Rule-only + repair loop	0.8926	0.6714	0.5988	0.42×	0.58×
Rule+Open-VLM + repair loop	0.9047	0.7062	0.7160	0.73×	0.87×
VideoGenDoctor-full	0.9110	0.7316	0.7639	1.00×	1.00×
Rule+GPT-4V + repair loop	0.9276	0.7688	0.7917	2.75×	2.31×
VLM structured report	0.8869	0.6639	0.6810	1.16×	1.20×

297 B Additional Experimental Tables

298 This appendix collects construction details, audit tables, stress analyses, and per-slice results that
299 support the main experimental claims without interrupting the main narrative.

Table 14: Operator-template alignment analysis for the expanded controlled fixture. The table makes explicit which observable cues, rule signals, failure codes, and repair-plan templates are aligned by construction.

Perturbation operator	Observable cue	Rule / feature signal	Failure code	Repair-plan template	Alignment type
Identity-anchor removal	face or body appearance drift	face embedding drift; character-region embedding drift	ID_FACE_DRIFT, ID_BODY_DRIFT	inject_reference_frame; replace_character_anchor	identity-anchor
Required-prop dropping	required object becomes absent or inconsistent	object detector absence; VLM prop-absence judgment	AL_PROP_MISSING	inject_prop	prop-presence
Camera-movement alteration	requested trajectory is replaced by static or wrong motion	global flow direction; trajectory mismatch	CA_MOVE_WRONG	set_camera_movement	motion-control
Shot-type / framing alteration	framing violates requested shot type	crop statistics; VLM framing judgment	CA_SHOT_TYPE_WRONG	adjust_framing	framing-control
Camera-shake injection	unstable camera with unintended jitter	flow instability; camera-shake heuristic	CA_SHAKE	stabilize_camera	stability-control
Action dropping / shifting	required event missing or misordered	action-track gap; VLM event-order judgment	MO_EVENT_MISSING	inject_action_keyframe; regenerate_segment	event-structure
Temporal jitter / frame-drop injection	discontinuity, duplicate frames, or rate mismatch	flow anomaly; near-zero motion plateau	MO_JITTER, MO_SEGMENT_BREAK	normalize_frame_rate; regenerate_segment	temporal-continuity
Compression re-encoding	blockiness and de-blocking artifacts	compression detector; texture degradation signal	ST_COMPRESSION_ARTIFACT	apply_enhancement	quality-restoration
Background-anchor weakening	scene background drifts across the shot	scene embedding drift; background-anchor mismatch	SC_BG_INCONSISTENCY	fix_background_anchor	scene-anchor

Table 15: Source composition of the expanded real-failure subset. The subset is failure-enriched: we generate 480 raw videos and retain 240 samples with annotator-verified visible failures. Sampling is balanced at 60 retained failures per generator, and generator-specific prompt or conditioning adapters are logged in the construction manifest. The subset is used only as an initial transfer check, not to estimate natural failure prevalence. We use publicly identifiable generator checkpoints to make the subset construction auditable and reproducible.

Generator	Version / checkpoint	Input type	#Prompts / specs	#Videos	Avg. duration	#Unique observed codes
CogVideoX	CogVideoX1.5-5B	text / ShotIR-to-prompt	60	60	8.4s	15
Wan	Wan2.1-T2V-14B	text / ShotIR-to-prompt	60	60	9.2s	16
Stable Video Diffusion	SVD-XT-1.1	image-to-video / reference frame	60	60	8.6s	14
HunyuanVideo	HunyuanVideo-1.5	text / multi-shot prompt	60	60	9.5s	17
Total	–	–	240	240	8.93s	18

Table 16: Real-failure and real-normal evaluation. The real-normal subset contains 120 videos retained from the same 480 raw generations but judged to contain no obvious failure. Whenever a diagnosis-oriented front end is listed, patch-related columns are computed after pairing that front end with the shared downstream repair loop. Best values are boldfaced; higher is better except for real-normal FPR and unnecessary patch rate, where lower is better.

Method	Real-failure Recall	Real-normal FPR	Unnecessary Patch Rate	No-op Accuracy	Specificity
Score-only	0.621	0.258	0.231	0.769	0.742
Rule-only	0.806	0.168	0.154	0.846	0.832
VLM structured report	0.818	0.217	0.205	0.795	0.783
VLM structured + self-consistency	0.837	0.167	0.150	0.850	0.833
VLM temporal proposal + patch	0.848	0.192	0.183	0.817	0.808
VideoGenDoctor-diagnosis	0.862	0.108	0.092	0.908	0.892
VideoGenDoctor-full	0.858	0.100	0.087	0.913	0.900

Table 17: Stress analysis on 72 ambiguous specification-violation cases. These samples contain possible but non-adjudicable deviations from the specification. Whenever a diagnosis-oriented front end is listed, patch-related columns are computed after pairing that front end with the shared downstream repair loop. Best values are boldfaced; higher is better for abstention-oriented columns, lower is better for false concrete patch, wrong-target patch, over-repair, and new artifacts.

Method	Abstention rate	Appropriate abstention	False concrete patch	Wrong-target patch	Over-repair	New artifacts
Score-only	0.292	0.361	0.514	0.222	0.347	0.083
VLM structured report	0.181	0.292	0.611	0.278	0.431	0.153
VLM temporal proposal + patch	0.222	0.347	0.556	0.236	0.389	0.194
VideoGenDoctor-diagnosis	0.500	0.639	0.292	0.125	0.208	0.069
VideoGenDoctor-full	0.472	0.625	0.250	0.111	0.181	0.111

Table 18: Stress analysis on 48 low-quality or non-adjudicable cases. These samples are dominated by global degradation, unclear content, or insufficient evidence for reliable code-level labels. Whenever a diagnosis-oriented front end is listed, patch-related columns are computed after pairing that front end with the shared downstream repair loop. Best values are boldfaced; higher is better for quality-gated abstention, lower is better for the remaining columns.

Method	Quality-gated abstention	False concrete patch	Wrong-target patch	Over-repair	New artifacts
Score-only	0.417	0.375	0.125	0.208	0.063
VLM structured report	0.292	0.458	0.188	0.271	0.104
VLM temporal proposal + patch	0.333	0.396	0.167	0.250	0.146
VideoGenDoctor-diagnosis	0.667	0.146	0.063	0.104	0.042
VideoGenDoctor-full	0.625	0.125	0.042	0.083	0.083

Table 19: Observed failure-code distribution in the expanded controlled fixture and real-failure subset. The full taxonomy contains 38 codes, but empirical claims are restricted to the observed codes below.

Failure code	Group	Controlled spans	Real-failure spans	Total spans
ID_FACE_DRIFT	Identity	150	112	262
ID_BODY_DRIFT	Identity	132	76	208
ID_CLOTHING_CHANGE	Identity	–	60	60
CA_MOVE_WRONG	Camera	240	104	344
CA_SHOT_TYPE_WRONG	Camera	144	84	228
CA_SHAKE	Camera	126	70	196
CA_WRONG_ANGLE	Camera	–	68	68
MO_JITTER	Motion	174	124	298
MO_FROZEN_FRAME	Motion	156	72	228
MO_SEGMENT_BREAK	Motion	162	88	250
MO_EVENT_MISSING	Motion	126	90	216
MO_ACTION_ORDER_WRONG	Motion	–	56	56
AL_PROP_MISSING	Alignment	198	116	314
AL_TEXT_PROMPT_MISMATCH	Alignment	–	132	132
ST_COMPRESSION_ARTIFACT	Style	252	80	332
ST_COLOR_SHIFT	Style	–	56	56
SC_BG_INCONSISTENCY	Scene	156	86	242
SC_OBJECT_TELEPORT	Scene	–	62	62
Total	–	2016	1536	3552

Table 20: Annotation reliability. Dual annotation is performed on a reliability subset before adjudication.

Subset	#Videos	#Candidate spans	#Annotators	Code agreement	Span agreement
Controlled reliability subset	72	468	2	$\kappa = 0.858$	tIoU = 0.716
Real-failure reliability subset	48	300	2	$\kappa = 0.803$	tIoU = 0.641
Combined	120	768	2	$\kappa = 0.831$	tIoU = 0.681

Table 21: Implementation details for VLM-based judges and external diagnosis/repair baselines.

Setting	Model	Frames	Res.	Prompt	Rel. cost
Rule+Open-VLM	Qwen2-VL-7B-Instruct	18	512px short side	structured QA	$0.34\times$
Rule+GPT-4V	GPT-4V endpoint	18	512px short side	structured QA	$2.75\times$
VLM free-form report	GPT-4V endpoint	12	512px short side	free-form critique	$1.12\times$
VLM structured report	GPT-4V endpoint	12	512px short side	JSON schema	$1.16\times$
VLM structured + self-consistency	GPT-4V endpoint	12 frames $\times$ 3 runs	512px short side	JSON schema, 3 runs	$3.48\times$
VLM temporal proposal + patch	GPT-4V endpoint	12	512px short side	span proposal + JSON	$2.05\times$
VLM-to-patch	GPT-4V endpoint	12	512px short side	JSON patch	$1.19\times$
Score+LLM-to-patch	scalar evaluator + LLM	0	–	score-to-patch JSON	N/A

Note. Most direct VLM baselines share the same hosted GPT-4V endpoint OpenAI [2023] to isolate prompt- and schema-level effects rather than backbone changes; Rule+Open-VLM provides an open-model reference. Hosted endpoint calls are logged with raw responses for auditability. ‘N/A’ denotes a scalar-evaluator-dominated pipeline whose cost is not directly comparable to frame-based VLM calls.

Table 22: Per-code reliability of the default VideoGenDoctor diagnosis configuration on the expanded controlled fixture.

Observed code	Precision	Recall	F1	tIoU@0.5
ID_FACE_DRIFT	0.938	0.914	0.926	0.748
ID_BODY_DRIFT	0.922	0.897	0.909	0.725
CA_MOVE_WRONG	0.932	0.916	0.924	0.734
CA_SHOT_TYPE_WRONG	0.913	0.889	0.901	0.705
CA_SHAKE	0.901	0.878	0.889	0.692
MO_JITTER	0.890	0.865	0.877	0.676
MO_FROZEN_FRAME	0.963	0.950	0.956	0.794
MO_SEGMENT_BREAK	0.906	0.883	0.894	0.704
MO_EVENT_MISSING	0.875	0.848	0.861	0.662
AL_PROP_MISSING	0.951	0.930	0.940	0.760
ST_COMPRESSION_ARTIFACT	0.950	0.966	0.958	0.779
SC_BG_INCONSISTENCY	0.910	0.885	0.897	0.686

Table 23: Per-code reliability on the expanded real-failure subset. The subset is more challenging than the controlled fixture, especially for motion, camera, and scene-level failures.

Failure code	#Spans	Precision	Recall	F1	tIoU@0.5
ID_FACE_DRIFT	112	0.884	0.848	0.866	0.648
ID_BODY_DRIFT	76	0.856	0.823	0.839	0.615
ID_CLOTHING_CHANGE	60	0.834	0.798	0.816	0.586
CA_MOVE_WRONG	104	0.866	0.832	0.849	0.628
CA_SHOT_TYPE_WRONG	84	0.838	0.803	0.820	0.590
CA_SHAKE	70	0.820	0.779	0.799	0.559
CA_WRONG_ANGLE	68	0.825	0.786	0.805	0.568
MO_JITTER	124	0.823	0.777	0.799	0.565
MO_FROZEN_FRAME	72	0.899	0.854	0.876	0.663
MO_SEGMENT_BREAK	88	0.827	0.792	0.809	0.582
MO_EVENT_MISSING	90	0.807	0.762	0.784	0.542
MO_ACTION_ORDER_WRONG	56	0.792	0.748	0.769	0.524
AL_PROP_MISSING	116	0.893	0.863	0.878	0.669
AL_TEXT_PROMPT_MISMATCH	132	0.842	0.794	0.817	0.596
ST_COMPRESSION_ARTIFACT	80	0.914	0.874	0.893	0.688
ST_COLOR_SHIFT	56	0.846	0.806	0.825	0.601
SC_BG_INCONSISTENCY	86	0.832	0.795	0.813	0.578
SC_OBJECT_TELEPORT	62	0.823	0.782	0.802	0.545

Table 24: Per-generator results on the real-failure and real-normal subsets. FPR is computed on 30 real-normal samples per generator.

Generator	Macro-F1	tIoU@0.5	Human Pass@2	Real-normal FPR	Main failure groups	Normal samples
CogVideoX	0.834	0.609	0.702	0.092	Camera, Motion, Alignment	30
Wan	0.821	0.588	0.681	0.108	Motion, Camera, Identity	30
Stable Video Diffusion	0.842	0.617	0.696	0.083	Identity, Style, Scene	30
HunyuanVideo	0.809	0.579	0.684	0.117	Camera, Motion, Alignment	30
Average	0.827	0.598	0.691	0.100	–	120

Table 25: Leave-one-perturbation-out evaluation. Each held-out operator is excluded from threshold tuning and then used only for testing.

Held-out perturbation operator	Macro-F1	tIoU@0.5	Top-3 hit
Identity-anchor removal	0.884	0.681	0.644
Required-prop dropping	0.902	0.704	0.672
Camera-movement alteration	0.871	0.653	0.612
Shot-type / framing alteration	0.862	0.641	0.606
Camera-shake injection	0.851	0.628	0.590
Action dropping / shifting	0.838	0.612	0.568
Temporal jitter / frame-drop injection	0.846	0.621	0.577
Compression re-encoding	0.910	0.719	0.688
Background-anchor weakening	0.868	0.646	0.610
Average	0.870	0.656	0.619

Table 26: Leave-one-source-group-out evaluation over the six controlled-fixture domain groups.

Held-out source group	Macro-F1	tIoU@0.5	Top-3 hit
Indoor / human-object	0.899	0.714	0.665
Outdoor / motion	0.887	0.702	0.651
Multi-object interaction	0.892	0.706	0.657
Camera-control scene	0.875	0.683	0.631
Style / lighting variation	0.907	0.722	0.676
Scene-change / background	0.884	0.694	0.642
Average	0.891	0.704	0.654

300 C Full Failure Taxonomy

301 Table 27 reports the machine-readable taxonomy used by VideoGenDoctor v0.1. The v0.1 taxonomy
302 contains 38 codes grouped into six categories. Each code is paired with an operational definition,
303 an evidence procedure, and a default patch template. The current benchmark evaluates the 12 codes
304 observed in VideoGenDoctor-Bench-v0, while the remaining codes define the extensible namespace
305 used by the report schema and patch compiler.

Table 27: Full VideoGenDoctor v0.1 failure taxonomy used by the report schema and patch compiler.

Code	Definition	Evidence Procedure	Patch Template
Identity			
ID_FACE_DRIFT	Face identity changes inconsistently between frames or segments.	Compare face embeddings across keyframes and flag large cosine distance.	inject_ reference_ frame
ID_BODY_DRIFT	Body appearance, clothing, or build changes inconsistently.	Compare character-region embeddings across segments.	inject_ reference_ frame
ID_WRONG_CHARACTER	A visible character does not match any specified character.	Check face embeddings against known character anchors.	replace_ character_ anchor

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Code	Definition	Evidence Procedure	Patch Template
ID_CLOTHING_CHANGE	Clothing changes between shots without narrative justification.	Measure character-region appearance drift across segments.	fix_character_consistency
ID_MULTIPLE_FACES	Multiple distinct faces appear when one character is expected.	Count distinct identities detected in a single-character shot.	adjust_character_count
ID_NO_FACE_VISIBLE	An expected face is absent when the shot requires face visibility.	Check face detection under close-up or medium-shot constraints.	adjust_framing
Scene			
SC_BG_INCONSISTENCY	The background changes unexpectedly between segments.	Measure background-region embedding drift across consecutive segments.	fix_background_anchor
SC_ENVIRONMENT_MISMATCH	The scene environment does not match the specified location.	Compare frames with the location description using image-text similarity.	set_environment
SC_LIGHTING_SHIFT	Lighting changes inconsistently within a shot.	Measure per-frame luminance variation within the segment.	normalize_lighting
SC_OBJECT_TELEPORT	A scene object appears or disappears without plausible motion.	Track object boxes across adjacent frames.	stabilize_scene_objects
SC_BG_FLICKER	The background flickers or pulses unnaturally.	Detect high-frequency luminance variation in background regions.	fix_background_anchor
SC_WRONG_TIME_OF_DAY	The time of day does not match the specification.	Classify sky and ambient lighting against the expected time.	set_time_of_day
Motion			
MO_JITTER	Unnatural frame-to-frame jitter appears outside intended camera motion.	Measure optical-flow magnitude variance and direction instability.	normalize_frame_rate
MO_FRAME_DROP	Duplicate or dropped frames produce visible stutter.	Detect near-zero adjacent-frame flow when motion is expected.	interpolate_frames
MO_UNNATURAL_MOTION	Character or object motion is physically implausible.	Compare flow direction and magnitude with expected motion constraints.	constrain_motion
MO_EVENT_MISSING	A required action or event does not occur in the expected segment.	Ask the VLM judge whether the specified action is visible.	inject_action_keyframe
MO_ACTION_ORDER_WRONG	Actions occur in the wrong temporal order.	Compare detected action timestamps with the specification order.	reorder_segments
MO_MOTION_BLUR_EXCESS	Excessive motion blur degrades keyframes.	Use sharpness measures such as Laplacian variance.	reduce_motion_blur
MO_FROZEN_FRAME	The video is frozen when motion is expected.	Detect near-zero flow across the full segment.	regenerate_segment
MO_SEGMENT_BREAK	A visual discontinuity occurs at a segment boundary.	Detect embedding-drift spikes at boundaries.	smooth_segment_transition
Camera			
CA_MOVE_WRONG	Camera movement deviates from the specified movement.	Compare optical-flow direction patterns with the expected movement type.	set_camera_movement
CA_SHOT_TYPE_WRONG	The shot type does not match the specification.	Compare face or body box size ratios with the expected shot type.	adjust_framing
CA_UNINTENDED_ZOOM	Unplanned zoom-in or zoom-out occurs.	Detect radial flow without a zoom specification.	remove_zoom
CA_SHAKE	Camera shake appears outside the specified movement.	Detect high-frequency oscillation in global optical flow.	stabilize_camera
CA_WRONG_ANGLE	Camera angle deviates from the specification.	Ask the VLM judge to compare the observed angle with the target angle.	set_camera_angle
CA_ROLL	Unintended camera roll appears.	Estimate the rotational component of optical flow.	remove_camera_roll
Alignment			
AL_PROP_MISSING	A required prop is not visible in the segment.	Use object detection or VLM verification for prop absence.	inject_prop
AL_PROP_WRONG_PLACEMENT	A required prop appears in the wrong region.	Compare the prop box with the expected screen region.	reposition_prop

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Code	Definition	Evidence Procedure	Patch Template
AL_TEXT_PROMPT_MISMATCH	The generated content does not align with the prompt.	Measure text-image similarity and verify mismatches with a judge.	strengthen_prompt_guidance
AL_CHARACTER_COUNT_WRONG	The number of visible characters does not match the specification.	Count faces or bodies and compare with the expected count.	adjust_character_count
AL_WRONG_PROP	A visible prop is not listed in the specification.	Detect high-confidence unlisted objects.	remove_prop
AL_PROP_OCCLUDED	A required prop is present but substantially occluded.	Estimate visible area and occlusion of the prop box.	reposition_prop
Style			
ST_COLOR_SHIFT	The color palette changes inconsistently across segments.	Measure HSV histogram distance between segments.	apply_color_grading
ST_ART_STYLE_INCONSISTENCY	The art style shifts between segments.	Compare style embeddings across segments.	apply_style_transfer
ST_COMPRESSION_ARTIFACT	Visible compression artifacts degrade video quality.	Detect blocking, ringing, or quality-score degradation.	apply_enhancement
ST_TEXTURE_FLICKER	Texture patterns flicker or shimmer unnaturally.	Measure high temporal frequency in texture regions.	smooth_texture
ST_OVEREXPOSURE	Significant frame regions are overexposed.	Count the fraction of saturated high-value pixels.	adjust_exposure
ST_UNDEREXPOSURE	Significant frame regions are underexposed.	Count the fraction of near-black pixels.	adjust_exposure

D Judge Protocol

The Stage-2 judge receives the top- K candidate segments produced by Stage-1. For each segment, the input includes the segment identifier, temporal bounds, keyframe paths, candidate failure codes, and specification context, including required props, expected actions, camera movement, shot type, and character anchors. The judge answers structured questions rather than producing a free-form report. This design keeps the output comparable across local open-source VLMs and hosted vision-capable models.

For identity failures, the judge checks whether the face, body appearance, clothing, or other distinctive visual attributes of the main character change inconsistently across keyframes. For alignment failures, it verifies whether required props are visible, whether they appear in the expected screen region, and whether the generated content matches the prompt or structured specification. For camera failures, it compares the observed camera movement, shot type, and viewpoint angle with the requested control signal. For motion failures, it verifies whether required actions occur, whether they occur in the specified order, and whether stutter, frozen frames, or segment breaks are visible. For style and scene failures, it checks lighting, color, compression artifacts, background consistency, and environment match.

The output is a structured object containing a natural-language answer for each question, a confidence score, per-code probability updates, and a re-ranking of keyframes. The final failure score is computed as

$$\tilde{\sigma}_{s,c} = (1 - \alpha)\sigma_{s,c}^{(1)} + \alpha\sigma_{s,c}^{(2)},$$

where $\sigma_{s,c}^{(1)}$ is the Stage-1 score, $\sigma_{s,c}^{(2)}$ is the judge score, and $\alpha = 0.6$ by default. All judge calls record the model name, prompt, raw response, token count when available, and output path to support reproducibility.

E Patch Template Examples

VideoGenDoctor compiles each diagnosis record into one or more patch actions. If a ShotIR specification is available, the compiler emits field-level diffs. If the generator does not expose a structured specification interface, the compiler emits generator-agnostic repair plans that can be translated into prompt edits, reference-frame injection, segment-level re-rendering, or post-processing steps.

```
{"action": "inject_reference_frame",
  "field": "character_id",
```

```

336 "value": "<anchor_frame>",
337 "reason": "ID_FACE_DRIFT detected at t=1.2s"}

338 {"action": "set_camera_movement",
339 "field": "camera_movement",
340 "value": "<expected_camera_movement>",
341 "reason": "CA_MOVE_WRONG detected in the localized span"}

342 {"action": "inject_prop",
343 "field": "props_required",
344 "value": "<missing_prop>",
345 "reason": "AL_PROP_MISSING verified by object or VLM evidence"}

346 {"action": "regenerate_segment",
347 "field": "segment",
348 "value": "<t0,t1>",
349 "reason": "MO_FROZEN_FRAME or MO_SEGMENT_BREAK is localized to this span"}

350 Each patch specifies an action type, the affected target, an optional update value, and a short rationale
351 tied to the evidence span. When multiple records affect the same target, the compiler groups them
352 before re-rendering to reduce conflicts among patch actions.

```

Table 28: Patch-to-generator adapter details. A repair plan is counted as adapter-executable only when it can be automatically translated into target-generator or editor inputs without manual rewriting. L0/L1 rows denote abstention or prompt-rewrite instructions rather than executable patches.

Repair-plan action	ShotIR field	Counted as	Adapter availability	Concrete generator/editor input	Fallback
inject_reference_frame	character.identity_anchor	L2	L2 available for SVD; L1 partial for text-to-video models	reference image input + identity prompt strengthening	prompt-only repair plan
set_camera_movement	camera.movement	L1	L1 partial	camera-control prompt rewrite; trajectory adapter when supported	no_op_uncertain if no camera control exists
set_camera_angle	camera.viewpoint	L1	L1 partial	viewpoint prompt rewrite; optional camera-control token	repair suggestion only
inject_prop	props_required	L1 / L2	L1 available; L2 partial with mask/reference input	prop phrase insertion + optional reference or mask	prompt edit
reposition_prop	props.region	L1	L1 partial	region-aware prompt rewrite or mask-conditioned edit	prompt-only repair plan
regenerate_segment	temporal span	L2 / L3	L2 available when segment rerendering is supported	segment-level rerender or temporal inpainting call	full-clip rerender
stabilize_camera	camera.constraint	L1 / L2	L1 partial	no-shake constraint or stabilization post-process	repair suggestion only
normalize_frame_rate	motion.continuity	L2	L2 partial	frame interpolation or temporal smoothing post-process	regenerate_segment
apply_enhancement	style.quality	L2	L2 available	enhancement / deblocking post-process	prompt edit for higher quality
fix_background_anchor	scene.background_anchor	L1	L1 partial	background-anchor prompt strengthening; optional reference image	prompt-only repair plan
no_op_uncertain	–	L0	L0 available	abstain from concrete repair	no action

353 In the reported Human Pass@K evaluation, the concretely executed actions are the L2/L3 subset
354 exposed by the active backend, primarily reference-frame injection, segment rerendering or temporal
355 smoothing, enhancement or deblocking, and mask- or reference-conditioned prop insertion when
356 supported. L0/L1 rows remain structured repair-plan instructions and are not counted as executable
357 patches.

358 F Annotation Guide Summary

359 Annotators review each perturbed video together with the original prompt or ShotIR specification
360 and the auto-assigned candidate failure codes. For each candidate, annotators determine whether
361 the failure is visually present, assign the tightest temporal span that contains the visible deviation,

and select representative keyframes when available. Annotators may also add missing failures if the perturbation produces an additional visible artifact.

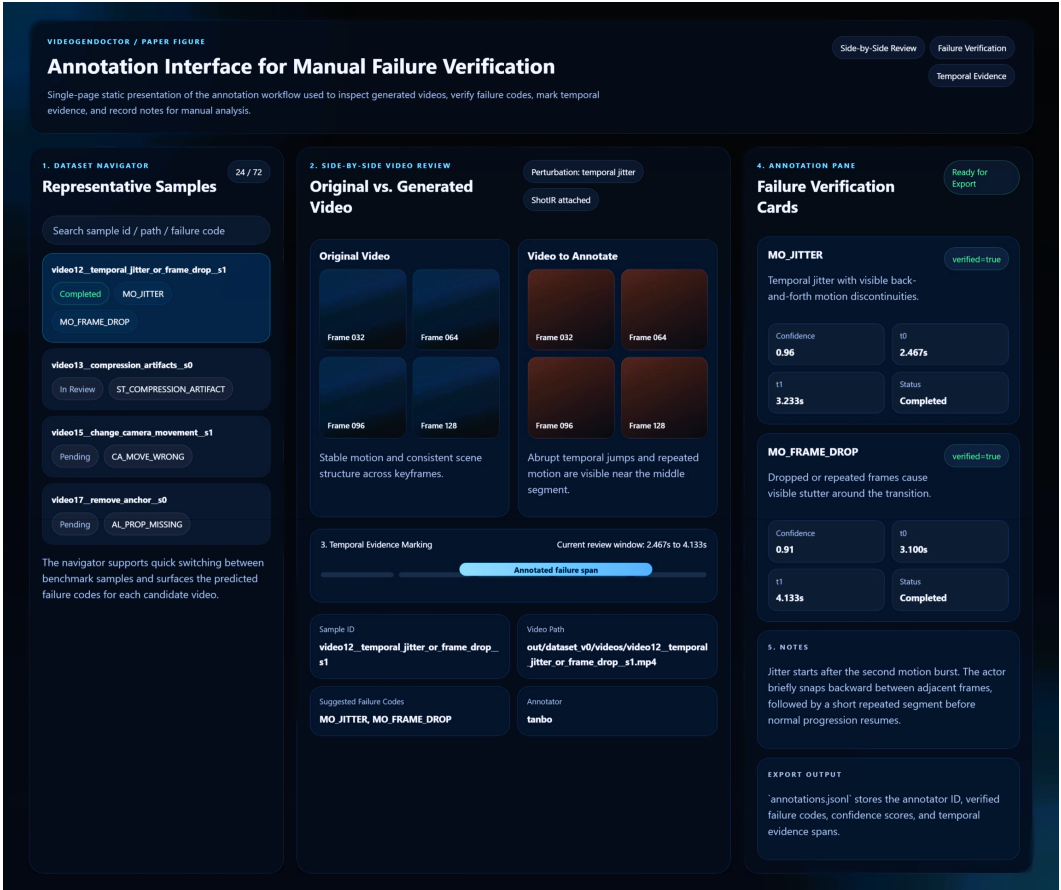


Figure 5: Annotation interface used for manual verification in VideoGenDoctor-Bench-v0. The tool exposes a sample navigator, side-by-side review of the original and perturbed videos, temporal evidence marking, structured failure verification cards, and free-form notes for adjudication.

The annotation record is stored as one JSON object per video and includes the video identifier, annotator identifier, verified failure codes, confidence values, evidence spans, and free-form notes. A failure is retained in the benchmark only when the annotator can verify it from the rendered video and specification, without relying on perturbation metadata alone. Ambiguous cases are handled conservatively: if a deviation is not visually identifiable, the candidate is marked as not verified.

The expanded experiment fixture includes a dual-annotated reliability subset of 120 videos and 768 candidate evidence spans. The subset contains both controlled perturbation samples and naturally occurring real-failure samples. Larger multi-annotator validation with more than two independent annotators remains future work.

373 G Example Diagnostic Report

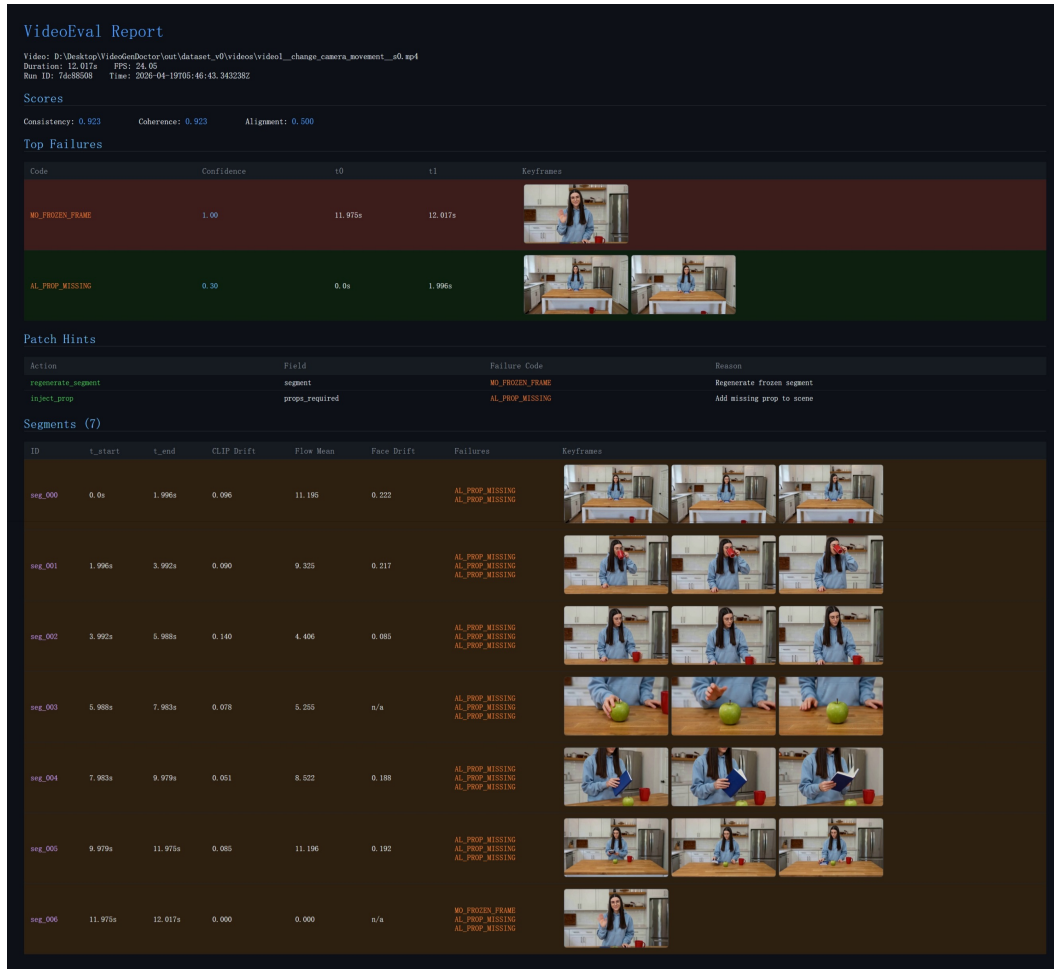


Figure 6: Example VideoGenDoctor diagnostic report for a perturbed video. The report summarizes global scores, ranked failure records with confidence and temporal spans, representative keyframes, patch hints derived from the retained failure codes, and segment-level evidence used for localized review and repair.

374 H Direct VLM Baseline Prompt Templates

375 The four direct VLM/LLM baselines differ only in their allowed information access and output schema.
 376 The free-form and structured VLM report baselines receive sampled frames and the specification.
 377 The VLM-to-patch baseline also receives sampled frames, the taxonomy names, and the repair-plan
 378 vocabulary, but not VideoGenDoctor’s rule scores or patch compiler outputs. The Score+LLM-to-
 379 patch baseline receives scalar evaluator scores and the specification only; it does not receive sampled
 380 frames. These information boundaries are fixed before evaluation and are used for all samples.

381 **F.1 Free-form VLM report prompt.** The model receives the video specification, sampled
 382 keyframes with timestamps, and a short instruction: identify visible failures, explain the evidence,
 383 and propose repair suggestions. The response is then parsed into taxonomy codes and evidence spans
 384 for evaluation.

385 **F.2 Structured VLM report prompt.** The model receives the same input but must output
 386 JSON objects with fields failure_code, temporal_span, affected_target, confidence,
 387 evidence_keyframes, rationale, and repair_action. Failures without visual evidence must
 388 be marked as uncertain rather than hallucinated.

```

389 You are evaluating whether a generated video follows the given specification.
390 For each visible failure, return a JSON record:
391 {
392     "failure_code": "<taxonomy code>",
393     "temporal_span": [t0, t1],
394     "affected_target": "<character / prop / camera / motion / style / scene>",
395     "confidence": <0-1>,
396     "evidence_keyframes": ["frame@time", ...],
397     "rationale": "short visual reason",
398     "repair_action": "<patch action>"
399 }
400 Only report failures that are visually supported by the frames.

```

F.3 VLM-to-patch prompt. The VLM-to-patch baseline receives the same video specification and sampled keyframes as the direct VLM diagnosis baselines, but it is asked to output repair actions directly rather than diagnosis records. This baseline is allowed to access the taxonomy names and the supported repair-plan vocabulary, but it is not given VideoGenDoctor’s Stage-1 rule scores, candidate segments, or patch compiler outputs. It may infer temporal spans from the sampled frames, but it does not receive ground-truth spans or VideoGenDoctor-predicted spans. This setting tests whether a VLM can directly produce structured repair plans from the specification and visual evidence.

```

408 You are a video-generation repair planner.
409
410 Input:
411 1. A video specification or ShotIR-style instruction.
412 2. Sampled keyframes from the generated video, each with a timestamp.
413 3. The failure-code taxonomy names.
414 4. The supported repair-plan vocabulary.
415
416 Your task:
417 Inspect the sampled frames and the specification. Identify only visually supported
418 problems and output structured repair-plan actions. Do not write a free-form critique.
419 Do not invent failures that are not visible in the frames.
420
421 You may use the following failure-code groups:
422 - Identity: ID_FACE_DRIFT, ID_BODY_DRIFT, ID_WRONG_CHARACTER,
423   ID_CLOTHING_CHANGE, ID_MULTIPLE_FACES, ID_NO_FACE_VISIBLE
424 - Camera: CA_MOVE_WRONG, CA_SHOT_TYPE_WRONG, CA_UNINTENDED_ZOOM,
425   CA_SHAKE, CA_WRONG_ANGLE, CA_ROLL
426 - Motion: MO_JITTER, MO_FRAME_DROP, MO_UNNATURAL_MOTION,
427   MO_EVENT_MISSING, MO_ACTION_ORDER_WRONG, MO_MOTION_BLUR_EXCESS,
428   MO_FROZEN_FRAME, MO_SEGMENT_BREAK
429 - Alignment: AL_PROP_MISSING, AL_PROP_WRONG_PLACEMENT,
430   AL_TEXT_PROMPT_MISMATCH, AL_CHARACTER_COUNT_WRONG,
431   AL_WRONG_PROP, AL_PROP_OCCLUDED
432 - Style: ST_COLOR_SHIFT, ST_ART_STYLE_INCONSISTENCY,
433   ST_COMPRESSION_ARTIFACT, ST_TEXTURE_FLICKER,
434   ST_OVEREXPOSURE, ST_UNDEREXPOSURE
435 - Scene: SC_BG_INCONSISTENCY, SC_ENVIRONMENT_MISMATCH,
436   SC_LIGHTING_SHIFT, SC_OBJECT_TELEPORT, SC_BG_FLICKER,
437   SC_WRONG_TIME_OF_DAY
438
439 Supported repair-plan vocabulary:
440 - inject_reference_frame
441 - replace_character_anchor
442 - fix_character_consistency
443 - adjust_framing
444 - set_camera_movement
445 - set_camera_angle

```

```

446 - stabilize_camera
447 - remove_zoom
448 - remove_camera_roll
449 - inject_prop
450 - reposition_prop
451 - remove_prop
452 - adjust_character_count
453 - strengthen_prompt_guidance
454 - inject_action_keyframe
455 - reorder_segments
456 - regenerate_segment
457 - interpolate_frames
458 - normalize_frame_rate
459 - constrain_motion
460 - smooth_segment_transition
461 - reduce_motion_blur
462 - apply_color_grading
463 - apply_style_transfer
464 - apply_enhancement
465 - smooth_texture
466 - adjust_exposure
467 - fix_background_anchor
468 - set_environment
469 - normalize_lighting
470 - stabilize_scene_objects
471 - set_time_of_day
472 - no_op_uncertain
473
474 Return a JSON object with the following schema:
475 {
476   "patches": [
477     {
478       "patch_id": "<unique patch id>",
479       "failure_code": "<taxonomy code or uncertain>",
480       "action": "<one item from the supported repair-plan vocabulary>",
481       "target": "<character / prop / camera / motion / style / scene target>",
482       "temporal_span": [<start_sec>, <end_sec>],
483       "evidence_keyframes": ["frame@time", "..."],
484       "confidence": <0-1>,
485       "patch_fields": {
486         "field": "<generator or ShotIR field to modify>",
487         "value": "<new value, constraint, or reference to insert>"
488       },
489       "rationale": "<short visual reason>"
490     }
491   ],
492   "global_notes": "<one-sentence summary>",
493   "uncertain": <true or false>
494 }
495
496 Rules:
497 1. Output valid JSON only.
498 2. Every patch must use an action from the supported repair-plan vocabulary.
499 3. If no visually supported failure is visible, return:
500   {"patches": [], "global_notes": "No visually supported repair action.", "uncertain": true}
501 4. If the temporal span is uncertain, estimate it from the nearest sampled frame
502    timestamps and set "confidence" below 0.5.
503 5. Do not output patches for purely aesthetic preferences unless the specification
504    explicitly requires them.

```

505 **F4 Score+LLM-to-patch prompt.** The Score+LLM-to-patch baseline receives scalar evaluator
 506 outputs and the original video specification, but it does not receive sampled frames. This baseline is
 507 intentionally weaker in visual grounding but represents a common production setting where low-level
 508 evaluators return scalar scores and an LLM converts low-scoring dimensions into repair suggestions.
 509 Because it cannot inspect frames, it is not allowed to claim frame-specific visual evidence. It may
 510 output approximate temporal spans only when the scalar evaluator provides segment-level scores;
 511 otherwise the temporal span must be marked as null. This makes the information boundary explicit
 512 and prevents unfair comparison with frame-based VLM baselines.

513 You are a repair planner for controllable video generation.

514

515 Input:

- 516 1. The original video specification or ShotIR-style instruction.
- 517 2. Scalar evaluator outputs, such as quality, alignment, motion, identity,
 518 camera, style, and scene-consistency scores.
- 519 3. Optional segment-level scalar scores if available.
- 520 4. The supported repair-plan vocabulary.

521

522 Important information boundary:

523 You do NOT receive sampled frames or the video itself. Therefore:

- 524 - Do not claim that a failure is visually observed.
- 525 - Do not mention specific visual evidence or keyframes.
- 526 - Only infer likely repair actions from low-scoring evaluator dimensions.
- 527 - Use "evidence_source": "scalar_score" for all patches.
- 528 - Use "temporal_span": null unless segment-level scores identify a specific interval.

529

530 Supported repair-plan vocabulary:

- 531 - inject_reference_frame
- 532 - replace_character_anchor
- 533 - fix_character_consistency
- 534 - adjust_framing
- 535 - set_camera_movement
- 536 - set_camera_angle
- 537 - stabilize_camera
- 538 - remove_zoom
- 539 - remove_camera_roll
- 540 - inject_prop
- 541 - reposition_prop
- 542 - remove_prop
- 543 - adjust_character_count
- 544 - strengthen_prompt_guidance
- 545 - inject_action_keyframe
- 546 - reorder_segments
- 547 - regenerate_segment
- 548 - interpolate_frames
- 549 - normalize_frame_rate
- 550 - constrain_motion
- 551 - smooth_segment_transition
- 552 - reduce_motion_blur
- 553 - apply_color_grading
- 554 - apply_style_transfer
- 555 - apply_enhancement
- 556 - smooth_texture
- 557 - adjust_exposure
- 558 - fix_background_anchor
- 559 - set_environment
- 560 - normalize_lighting
- 561 - stabilize_scene_objects
- 562 - set_time_of_day


```

563 - no_op_uncertain
564
565 Return a JSON object with the following schema:
566 {
567   "patches": [
568     {
569       "patch_id": "<unique patch id>",
570       "score_dimension": "<low-scoring evaluator dimension>",
571       "inferred_failure_type": "<short inferred failure description>",
572       "action": "<one item from the supported repair-plan vocabulary>",
573       "target": "<character / prop / camera / motion / style / scene target>",
574       "temporal_span": null,
575       "evidence_source": "scalar_score",
576       "triggering_scores": {
577         "<score_name>": <score_value>
578       },
579       "confidence": <0-1>,
580       "patch_fields": {
581         "field": "<generator or ShotIR field to modify>",
582         "value": "<new value, constraint, or reference to insert>"
583       },
584       "rationale": "<short reason based only on scalar scores and specification>"
585     }
586   ],
587   "global_notes": "<one-sentence summary>",
588   "uncertain": <true or false>
589 }
590
591 Rules:
592 1. Output valid JSON only.
593 2. Do not claim visual evidence because frames are not provided.
594 3. Do not output frame IDs or evidence keyframes.
595 4. Use temporal_span = null unless segment-level scalar scores are provided.
596 5. Every patch must use an action from the supported repair-plan vocabulary.
597 6. If the scalar scores do not indicate a clear repair target, return:
598 {"patches": [], "global_notes": "No reliable score-based repair action.", "uncertain": true}

```

599 **F.5 Parsing and invalid-output handling.** Invalid JSON outputs are repaired using a deterministic
600 parser when possible. If the output cannot be parsed, the prediction is marked invalid. Failure
601 codes outside the taxonomy are mapped to the nearest taxonomy code only when an exact synonym
602 exists; otherwise they are counted as false positives. Predictions without temporal spans receive
603 no localization credit. Hallucinated failures without visual evidence are counted as false positives.
604 Repair actions outside the supported repair-plan vocabulary are marked invalid for repair-plan scoring
605 and non-adaptor-executable for adaptor scoring.

606 I Blind Human Repair Evaluation Protocol

607 Raters are shown the specification and one output video, without method names or diagnosis records.
608 They answer whether the video satisfies the specification overall, whether the target failure is resolved,
609 whether the repair introduces new artifacts, and how useful the patch is on a 1–5 scale. Human
610 Pass@ K is the fraction of samples judged successful after at most K repair attempts.

Table 29: Blind human repair evaluation rubric for patch usefulness.

Score	Criterion
1	The patch does not address the target failure.
2	The patch is related to the failure but is not actionable or is largely insufficient.
3	The patch is actionable but incomplete or requires substantial manual interpretation.
4	The patch directly addresses the target failure with minor ambiguity.
5	The patch directly resolves the target failure and preserves surrounding valid content.

611 A repaired video is marked as pass if the target failure is resolved and no severe new artifact is
612 introduced. Human Pass@K is the fraction of samples judged successful after at most K repair
613 attempts. New artifacts are marked when raters observe a visible degradation introduced by repair that
614 was not present in the original generated video. Each repaired video is rated by 3 independent raters.
615 We report the majority-vote pass/fail label, the mean patch-usefulness score, and the majority-vote
616 new-artifact label.

Table 30: Error analysis on ambiguous and real-normal samples. The table focuses on no-op behavior and wrong-target repair planning. Best values are boldfaced for interpretable comparison columns; the no_op_uncertain rate is reported descriptively rather than bold-ranked.

Method	no_op_uncertain rate	Correct no-op rate	Missed necessary repair	Wrong-target patch	Over-confident patch
Score-only	0.286	0.524	0.119	0.171	0.321
VLM structured report	0.219	0.476	0.096	0.229	0.396
VLM temporal proposal + patch	0.257	0.518	0.108	0.200	0.350
VideoGenDoctor-diagnosis	0.552	0.738	0.142	0.095	0.181
VideoGenDoctor-full	0.514	0.724	0.156	0.081	0.164

Table 31: Qualitative case-study summary. Cases are sampled from controlled, real-failure, and ambiguous stress subsets.

Case type	#Cases	Majority-judged successful / appropriate	Typical outcome
Successful structured repair plan	24	18	localized repair plan resolves target failure
Ambiguous case with abstention	24	16	no_op_uncertain avoids over-repair
Over-repair / wrong-target failure	24	7	concrete patch targets wrong object or span

617 J Bootstrap Confidence Intervals

618 We use 10,000 video-level bootstrap resamples and report percentile 95% confidence intervals. For the
619 controlled fixture, resampling is performed over 324 videos. For the real-failure subset, resampling is
620 performed over 240 videos. For blind human repair evaluation, resampling is performed over the
621 stratified human-rated subset. For paired comparisons, we resample videos with replacement and
622 compute the distribution of metric differences between two methods. Small point-estimate differences
623 are not treated as meaningful unless the paired confidence interval excludes zero.

Table 32: Leave-one-code-family-out rule ablation. For the held-out family, dedicated family-specific rules are disabled; only generic embedding drift, optical-flow anomaly, scene-change peaks, and structured VLM verification are retained.

Held-out family	Macro-F1	tIoU@0.5	Top-3 hit	Main degradation source
Identity	0.782	0.579	0.536	face/body-specific anchors removed
Camera	0.754	0.546	0.501	camera-flow templates disabled
Motion	0.721	0.512	0.468	jitter / segment-break rules disabled
Alignment	0.801	0.603	0.558	prop detector and placement rules disabled
Style	0.836	0.641	0.590	compression/style detectors disabled
Scene	0.748	0.528	0.487	background-anchor rules disabled
Average	0.774	0.568	0.523	–

The large degradation under leave-one-code-family-out ablation confirms that VideoGenDoctor should not be interpreted as an unconstrained open-ended failure discovery system. The experiment clarifies which portions of the performance depend on family-specific detectors and which portions remain supported by generic evidence signals and VLM verification.

K Real-Failure Construction Manifest

Table 33: Generator configuration metadata for the real-failure and real-normal subsets. Hosted or version-changing endpoints are logged with access date and endpoint identifier.

Generator	Checkpoint / endpoint	fps	Frames	Resolution	Steps	Guidance	Scheduler	Seed range	Reference source / prompt rule
CogVideoX	CogVideoX1.5-5B	16	128	768×432	50	6.0	DPMSolver	10000–10059	ShotIR-to-prompt template v1
Wan	Wan2.1-T2V-14B	16	128	768×432	50	5.0	FlowMatchEuler	18000–18059	ShotIR-to-prompt template v1
Stable Video Diffusion	SVD-XT-1.1	14	112	768×432	40	7.5	EulerDiscrete	26000–26059	reference frame from ShotIR anchor
HunyuanVideo	HunyuanVideo-1.5	16	128	768×432	50	7.0	FlowMatchEuler	34000–34059	multi-shot prompt template v1

When a generator does not expose one of these configuration fields, the corresponding manifest value is set to null rather than omitted. For hosted or version-changing endpoints, we log the endpoint identifier, access date, raw request payload, and raw response metadata.

To make the real-failure subset auditable and reproducible, each generated video is accompanied by a JSON manifest. The manifest records the generator identity, checkpoint, input specification, generation configuration, output location, screening decision, verified failure codes, temporal evidence spans, and annotator identifiers. This protocol turns the real-failure subset from an informal collection of generated examples into a traceable evaluation resource.

Each manifest entry contains the following fields: `generator`, `checkpoint`, `prompt_id`, `input_type`, `shotir_spec`, `shotir_to_prompt_converted_text`, `reference_frame_id`, `seed`, `resolution`, `fps`, `num_frames`, `sampling_steps`, `guidance_scale`, `output_path`, `screening_status`, `verified_codes`, `evidence_spans`, and `annotator_ids`. The `screening_status` field takes one of four values: `generated`, `retained_candidate`, `verified`, or `excluded_ambiguous`. Only entries marked as `verified` are included in the reported real-failure evaluation.

```
{
  "video_id": "real_wan_00037",
  "generator": "Wan",
  "checkpoint": "Wan2.1-T2V-14B",
  "prompt_id": "shotir_prompt_037",
  "input_type": "text / ShotIR-to-prompt",
  "shotir_spec": {
    "shot_id": "S037",
    "duration_sec": 8.0,
    "characters": [
      {
        "id": "person_A",
        "description": "young man wearing a blue hoodie",
        "identity_anchor": "ref_person_A_004"
      }
    ],
    "required_props": [
      {
        "id": "red_water_bottle",
        "description": "red water bottle held in the right hand",
        "expected_presence": "visible throughout the shot"
      }
    ],
    "camera": {
      "shot_type": "medium shot",
      "movement": "slow left-to-right tracking",

```

```

670     "viewpoint": "front three-quarter view"
671 },
672 "action_order": [
673     "person_A enters the park path",
674     "person_A jogs while holding the red water bottle",
675     "person_A slows down near the bench"
676 ],
677 "style": {
678     "lighting": "natural daylight",
679     "visual_style": "realistic video"
680 },
681 "scene": {
682     "location": "urban park path",
683     "background_anchor": "trees and bench remain stable"
684 }
685 },
686 "shotir_to_prompt_converted_text": "A realistic 8-second medium shot of a young man wearing a k
687 "reference_frame_id": null,
688 "seed": 18473,
689 "resolution": "768x432",
690 "fps": 16,
691 "num_frames": 128,
692 "sampling_steps": 50,
693 "guidance_scale": 7.5,
694 "output_path": "real_failures/wan/real_wan_00037.mp4",
695 "screening_status": "verified",
696 "verified_codes": [
697     "AL_PROP_MISSING",
698     "CA_MOVE_WRONG",
699     "MO_SEGMENT_BREAK"
700 ],
701 "evidence_spans": [
702     {
703         "failure_code": "AL_PROP_MISSING",
704         "target": "red_water_bottle",
705         "start_sec": 3.25,
706         "end_sec": 6.75,
707         "evidence_keyframes": [
708             "frame_052@3.25s",
709             "frame_080@5.00s",
710             "frame_108@6.75s"
711         ],
712         "annotator_confidence": 0.91,
713         "notes": "The specified red water bottle is not visible after the character begins jogging
714     },
715     {
716         "failure_code": "CA_MOVE_WRONG",
717         "target": "camera_movement",
718         "start_sec": 0.00,
719         "end_sec": 4.00,
720         "evidence_keyframes": [
721             "frame_000@0.00s",
722             "frame_032@2.00s",
723             "frame_064@4.00s"
724         ],
725         "annotator_confidence": 0.84,
726         "notes": "The camera remains mostly static rather than performing the requested left-to-right
727     },
728     {

```

```

729     "failure_code": "MO_SEGMENT_BREAK",
730     "target": "temporal_continuity",
731     "start_sec": 5.50,
732     "end_sec": 5.94,
733     "evidence_keyframes": [
734         "frame_088@5.50s",
735         "frame_095@5.94s"
736     ],
737     "annotator_confidence": 0.78,
738     "notes": "A visible discontinuity occurs near the transition to the final action."
739 }
740 ],
741 "annotator_ids": [
742     "ann_02",
743     "ann_05"
744 ]
745 }

```

746 For privacy and release safety, file paths may be anonymized, but the manifest must preserve all fields
747 required to reproduce the generation configuration, reconstruct the ShotIR-to-prompt conversion, and
748 audit the verified evidence spans. When a generator does not expose a configuration field such as
749 guidance_scale, the field is set to null rather than omitted.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

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